

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                      |                                                 |
|----------------------|-------------------------------------------------|
| Product Name         | <u>Trichloroacetic acid</u>                     |
| CAS No               | 76-03-9                                         |
| Synonyms             | TCA                                             |
| Molecular Formula    | C <sub>2</sub> H Cl <sub>3</sub> O <sub>2</sub> |
| Molecular Weight     | 163.39                                          |
| Recommended Use      | Laboratory chemicals.                           |
| Uses advised against | No Information available                        |

|                         |                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------|
| Product Code            | <b>22156</b>                                                                                |
| Address                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| Emergency Tel.          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| Telephone / Fax Numbers | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| E-mail address          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

HSNO Approval Number     HSR002720

### GHS Classification

#### Physical hazards

Substances/mixtures corrosive to metal     Category 1

#### Health hazards

|                                                    |              |
|----------------------------------------------------|--------------|
| Acute Oral Toxicity                                | Category 4   |
| Skin Corrosion/Irritation                          | Category 1 A |
| Serious Eye Damage/Eye Irritation                  | Category 1   |
| Specific target organ toxicity - (single exposure) | Category 3   |

#### Environmental hazards

|                          |            |
|--------------------------|------------|
| Acute aquatic toxicity   | Category 1 |
| Chronic aquatic toxicity | Category 1 |

**Label Elements****Signal Word****Danger****Hazard Statements**

H314 - Causes severe skin burns and eye damage  
 H290 - May be corrosive to metals  
 H335 - May cause respiratory irritation  
 H410 - Very toxic to aquatic life with long lasting effects  
 H302 - Harmful if swallowed

**Precautionary Statements****Prevention**

P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
 P264 - Wash face, hands and any exposed skin thoroughly after handling  
 P201 - Obtain special instructions before use  
 P202 - Do not handle until all safety precautions have been read and understood  
 P271 - Use only outdoors or in a well-ventilated area  
 P280 - Wear protective gloves/protective clothing/eye protection/face protection  
 P273 - Avoid release to the environment

**Response**

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
 P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
 P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
 P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
 P310 - Immediately call a POISON CENTER or doctor  
 P363 - Wash contaminated clothing before reuse  
 P391 - Collect spillage

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

Toxicity to Soil Dwelling Organisms  
 Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component            | CAS No  | Weight % |
|----------------------|---------|----------|
| Trichloroacetic acid | 76-03-9 | >95      |

## Section 4 - First Aid Measures

**Description of first aid measures****General Advice**

Immediate medical attention is required. Show this safety data sheet to the doctor in

|                                            |                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | attendance.                                                                                                                                                                                                                                                                                                                                  |
| <b>New Zealand Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                                                                                                                                                                                                                                                   |
| <b>Inhalation</b>                          | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician or poison control center immediately. If not breathing, give artificial respiration. |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.                                                                                                                                                                          |
| <b>Skin Contact</b>                        | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.                                                                                                                                                                                                          |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Immediate medical attention is required. Never give anything by mouth to an unconscious person. Drink plenty of water.                                                                                                                                                                                               |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                             |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                         |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation                                              |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                                                                       |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

### Extinguishing media which must not be used for safety reasons

No information available.

### Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes. Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Chloroform, Carbon dioxide (CO<sub>2</sub>), Phosgene, Thermal decomposition can lead to release of irritating gases and vapors, Hydrogen chloride gas.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

#### Emergency procedures

Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.

#### Environmental Precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

**Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling****Advice on safe handling**

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Conditions for Safe Storage, Including any Incompatibilities****Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

**Incompatible Materials**

Strong oxidizing agents. Bases. Metals.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Control parameters****Exposure limits**

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

| Component            | New Zealand WEL                          | Australia                                | ACGIH TLV    | The United Kingdom |
|----------------------|------------------------------------------|------------------------------------------|--------------|--------------------|
| Trichloroacetic acid | TWA: 1 ppm<br>TWA: 6.7 mg/m <sup>3</sup> | TWA: 1 ppm<br>TWA: 6.7 mg/m <sup>3</sup> | TWA: 0.5 ppm |                    |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Appropriate engineering controls****Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Individual protection measures, such as personal protective equipment**

|                        |                                                                                                    |
|------------------------|----------------------------------------------------------------------------------------------------|
| <b>Eye Protection</b>  | Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications) |
| <b>Hand Protection</b> | Protective gloves                                                                                  |

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-------------------|-----------------|-----------------|-----------------------|
| Butyl rubber.  | > 480 minutes     | 0.7 mm          | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:** Particulates filter conforming to EN 143 (or AUS/NZ equivalent)  
**Recommended half mask:-** Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                 |                                          |
|------------------------------------------------|---------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Solid                           |                                          |
| <b>Appearance</b>                              | White                           |                                          |
| <b>Odor</b>                                    | of vinegar                      |                                          |
| <b>Odor Threshold</b>                          | No data available               |                                          |
| <b>pH</b>                                      | 1.2                             | (0.1M)                                   |
| <b>Melting Point/Range</b>                     | 52 - 58 °C / 125.6 - 136.4 °F   |                                          |
| <b>Softening Point</b>                         | No data available               |                                          |
| <b>Boiling Point/Range</b>                     | 196 °C / 384.8 °F               | @ 760 mmHg                               |
| <b>Flammability (liquid)</b>                   | Not applicable                  | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available        |                                          |
| <b>Explosion Limits</b>                        | No data available               |                                          |
| <b>Flash Point</b>                             | No information available        | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available               |                                          |
| <b>Decomposition Temperature</b>               | No data available               |                                          |
| <b>Viscosity</b>                               | Not applicable                  | Solid                                    |
| <b>Water Solubility</b>                        | 120 g/100 mL (20°C)             |                                          |
| <b>Solubility in other solvents</b>            | No information available        |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                 |                                          |
| <b>Component</b>                               | <b>log Pow</b>                  |                                          |
| Trichloroacetic acid                           | 1.44                            |                                          |
| <b>Vapor Pressure</b>                          | 1.2 mbar @ 50°C, 0.08 mbar @25C |                                          |
| <b>Density / Specific Gravity</b>              | 1.620                           |                                          |
| <b>Bulk Density</b>                            | No data available               |                                          |
| <b>Vapor Density</b>                           | Not applicable                  | Solid                                    |
| <b>Particle characteristics</b>                | No data available               |                                          |

**Other information**

Molecular Formula C<sub>2</sub> H Cl<sub>3</sub> O<sub>2</sub>  
Molecular Weight 163.39  
Evaporation Rate Not applicable - Solid

## Section 10 - Stability and Reactivity

**Reactivity** None known, based on information available

**Stability** Stable under normal conditions.

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

**Hazardous Polymerization** Hazardous polymerization does not occur.

**Hazardous Reactions** None under normal processing.

**Conditions to Avoid** Incompatible products, Excess heat.

**Incompatible Materials** Strong oxidizing agents, Bases, Metals.

**Hazardous Decomposition Products** Chloroform. Carbon dioxide (CO<sub>2</sub>). Phosgene. Thermal decomposition can lead to release of irritating gases and vapors. Hydrogen chloride gas.

## Section 11 - Toxicological Information

**Acute Effects****Information on likely routes of exposure****Product Information**

**Inhalation** Causes severe burns. May be harmful if inhaled. Harmful by inhalation.

**Eyes** Causes severe burns. Causes burns. Corrosive to the eyes and may cause severe damage including blindness. Risk of serious damage to eyes.

**Skin** Causes severe burns. May be harmful in contact with skin. Causes burns.

**Ingestion** Causes severe burns. May be harmful if swallowed. Ingestion causes burns of the upper digestive and respiratory tracts. Can burn mouth, throat, and stomach. Harmful if swallowed.

**Numerical measures of toxicity****(a) acute toxicity;**

**Oral** Based on available data, the classification criteria are not met

**Dermal** No data available

**Inhalation** No data available

| Component            | LD50 Oral      | LD50 Dermal               | LC50 Inhalation |
|----------------------|----------------|---------------------------|-----------------|
| Trichloroacetic acid | 3320 mg/kg rat | LD50 > 2000 mg/kg ( Rat ) |                 |

**(b) skin corrosion/irritation;** Category 1 A

**(c) serious eye damage/irritation;** Category 1

**(d) respiratory or skin sensitization;**

**Respiratory** No data available  
**Skin** No data available

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;**

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component            | New Zealand          | Australia | New South Wales | Western Australia | IARC     | EU | UK | Germany |
|----------------------|----------------------|-----------|-----------------|-------------------|----------|----|----|---------|
| Trichloroacetic acid | Suspected carcinogen |           |                 |                   | Group 2B |    |    |         |

**(g) reproductive toxicity;** No data available

**(h) STOT-single exposure;** Category 3

**Results / Target organs** Respiratory system

**(i) STOT-repeated exposure;** Based on available data, the classification criteria are not met

**Test method** Chronic Toxicity  
**Test species / Duration** dog / 90 days  
**Study result** NOEL = 26 mg/kg/d  
**Route of exposure** Oral  
**Target Organs** None known.

**(j) aspiration hazard;** Not applicable  
Solid

**Symptoms / effects, both acute and delayed**

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

## Section 12 - Ecological Information

**Ecotoxicity****Aquatic ecotoxicity**

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component            | Freshwater Fish | Water Flea | Freshwater Algae | Microtox |
|----------------------|-----------------|------------|------------------|----------|
| Trichloroacetic acid | >277 mg/l       | 110 mg/l   | 0.27 mg/l        |          |

**Terrestrial ecotoxicity**

| Component            | Earthworm                                                                                                                                                              | Avian | Honeybees |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------|
| Trichloroacetic acid | Acute toxicity: LC50 = 0.0964 mg/cm <sup>2</sup> (Eisenia foetida, 48 h, filter paper) Acute toxicity: LC50 = 1139.9 mg/kg (Eisenia foetida, 14 Days, soil dry weight) |       |           |

**Persistence and Degradability** Not readily biodegradable

**Persistence** Soluble in water, Persistence is unlikely, based on information available.

**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** Product has a low potential to bioconcentrate Bioaccumulation is unlikely

| Component            | log Pow | Bioconcentration factor (BCF) |
|----------------------|---------|-------------------------------|
| Trichloroacetic acid | 1,44    | 0.4-1.7 Cyprinus caprio       |

**Mobility** The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

**Other adverse effects**

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste treatment methods**

**Waste from Residues/Unused Products** Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Solutions with low pH-value must be neutralized before discharge. Do not let this chemical enter the environment.

## Section 14 - Transport Information

| Component                               | Hazchem Code |
|-----------------------------------------|--------------|
| Trichloroacetic acid<br>76-03-9 ( >95 ) | 2X           |

**NZS 5433:2020**

**UN-No** UN1839  
**Proper Shipping Name** Trichloroacetic acid, solid  
**Hazard Class** 8  
**Packing Group** II

**IATA**

**UN-No** UN1839  
**Proper Shipping Name** Trichloroacetic acid  
**Hazard Class** 8  
**Packing Group** II

**IMDG/IMO**



|                                                                                 |                                                                                                                         |
|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>UN-No</b>                                                                    | UN1839                                                                                                                  |
| <b>Proper Shipping Name</b>                                                     | Trichloroacetic acid, solid                                                                                             |
| <b>Hazard Class</b>                                                             | 8                                                                                                                       |
| <b>Packing Group</b>                                                            | II                                                                                                                      |
| <b>Environmental hazards</b>                                                    | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO                |
| <b>Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code</b> | Not applicable, packaged goods                                                                                          |
| <b>Special Precautions</b>                                                      | No special precautions required. Please refer to the applicable dangerous goods regulations for additional information. |
| <b>Additional information</b>                                                   | None known                                                                                                              |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                             |           |
|-----------------------------|-----------|
| <b>HSNO Approval Number</b> | HSR002720 |
|-----------------------------|-----------|

#### National Regulations

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### Certified handlers, tracking and controlled substance license requirements

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

| Component            | New Zealand          |
|----------------------|----------------------|
| Trichloroacetic acid | Suspected carcinogen |

### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

#### Authorisation/Restrictions according to EU REACH

| Component            | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Trichloroacetic acid | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

### International Inventories

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component            | CAS No  | NZIoC | AICS | EINECS | ELINCS | NLP | KECL     | IECSC | TCSI |
|----------------------|---------|-------|------|--------|--------|-----|----------|-------|------|
| Trichloroacetic acid | 76-03-9 | X     | X    | -      | -      | -   | KE-34058 | X     | X    |

| Component            | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|----------------------|---------|------|-----------------------------------------------|-----|------|-------|------|------|
| Trichloroacetic acid | 76-03-9 | X    | ACTIVE                                        | X   | -    | X     | X    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

### Legend

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**NZS 5433:2020** - Transport of Dangerous Goods on Land

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**AICS** - Australian Inventory of Chemical Substances

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists

**PNEC** - Predicted No Effect Concentration

**OECD** - Organisation for Economic Co-operation and Development

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** - Australian Code for the Transport of Dangerous Goods by Road and Rail

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

### Training Advice

Chemical incident response training.

### Revision Date

13-Mar-2023

### Revision Summary

Not applicable

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

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**End of Safety Data Sheet**